DEBUGGING PROMPTS: BEFORE–AFTER ANALYSIS AND IMPROVEMENTS

Day 1 - Assignment 2 – Generative AI

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1. Introduction

This assignment focuses on analyzing and improving poorly designed prompts used in customer support scenarios. The objective is to identify common issues in prompt design and enhance them using structured techniques.

Each case follows a three-step methodology:

- **Identification** – Recognizing flaws in the original prompt
- **Improvement** – Applying corrections to enhance clarity and effectiveness
- **Validation** – Demonstrating improved output quality

The prompts are evaluated based on real-world usability, clarity, and correctness.

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2. Debugging Cases

Case 1: Lack of Context and Structure

Category: Order Status

Original Prompt:
Answer the customer's question about their order.

Issues Identified:

- No role definition provided
- No order-related data included
- No structure for the response
- No tone or empathy guidance

Improved Prompt:
You are a customer support assistant for a company. A customer is asking about their order status.

Provide the following details:

- Current status of the order

- Expected delivery date
- Tracking information (if available)
- Offer further assistance

Use a polite and professional tone. Keep the response clear and concise.

Improved Response (Example):
Thank you for reaching out. Your order is currently in transit and is expected to arrive within the next few days. You can track your shipment using the tracking link provided. Please let us know if you need further assistance.

Explanation of Improvement:
The revised prompt introduces role clarity, structured output, and relevant data expectations. This ensures the response is informative, consistent, and user-friendly.

Case 2: Overconfidence and No Escalation

Category: Troubleshooting

Original Prompt:
Help the customer fix their issue and always provide a solution.

Issues Identified:

- Forces the system to give incorrect answers
- No product or issue details
- No escalation option
- No structured troubleshooting steps

Improved Prompt:
You are a technical support assistant. A customer is facing an issue with a product.

Provide troubleshooting steps in the following order:

1. Basic checks (restart, connection)
2. Intermediate fixes (update, reinstall)
3. Advanced steps (reset if necessary)

If the issue is not resolved, suggest contacting support for further assistance.

Improved Response (Example):
Please try restarting the device and checking your connections. If the issue persists, update the application or reinstall it. If none of these steps resolve the issue, please contact our support team for further assistance.

Explanation of Improvement:
The improved prompt removes unrealistic expectations and introduces a structured troubleshooting approach along with an escalation path.

Case 3: Missing Policy Logic

Category: Refund

Original Prompt:
Process this refund request.

Issues Identified:

- No refund policy mentioned
- No purchase details
- No decision-making logic
- No empathy in response

Improved Prompt:
You are a customer support agent handling a refund request.

Check whether the request falls within the refund policy period.

- If eligible: approve the refund and provide timeline
- If not eligible: explain politely and offer alternatives

Maintain a polite and empathetic tone.

Improved Response (Example):
Thank you for your request. Based on our policy, your purchase falls within the refund period, and we have initiated your refund. The amount will be credited within 5 to 7 business days.

Explanation of Improvement:
The revised prompt introduces decision logic and policy awareness, ensuring accurate and fair responses.

Case 4: Security Risk in Prompt

Category: Account / Login

Original Prompt:
Ask the customer for their username and password to verify their account.

Issues Identified:

- Serious security violation
- Encourages sharing sensitive information
- No secure verification method
- No compliance with standard policies

Improved Prompt:
You are a secure customer support assistant.

Help the user regain access to their account using safe methods such as:

- Password reset link
- OTP verification
- Registered email confirmation

Do not ask for sensitive information like passwords. Maintain a secure and professional tone.

Improved Response (Example):
You can reset your password using the “Forgot Password” option. A secure link will be sent to your registered email. Please avoid sharing your password with anyone.

Explanation of Improvement:
The improved prompt removes security risks and follows safe authentication practices, making it compliant and reliable.

Case 5: Vague Instructions and No Structure

Category: General Support

Original Prompt:
Be nice and help the customer.

Issues Identified:

- Vague instruction
- No clear goal
- No structure for response
- No actionable output

Improved Prompt:
You are a customer support specialist.

Respond to the customer by:

- Greeting them politely
- Understanding the issue
- Providing a clear solution
- Giving next steps or timeline

Use a professional and helpful tone.

Improved Response (Example):
Hello, thank you for contacting us. We understand your concern and are here to help. Based on your issue, please follow the steps provided. If the issue continues, we will assist you further.

Explanation of Improvement:
The improved version replaces vague instructions with a structured format, ensuring clarity and usefulness.

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3. Summary of Improvements

Case	Issue Type	Improvement Applied	Risk Level
Case 1	Lack of context	Added role and structure	Medium
Case 2	Overconfidence	Added step-by-step and escalation	High
Case 3	Missing logic	Introduced policy-based decisions	High
Case 4	Security flaw	Removed sensitive data request	Critical
Case 5	Vague instruction	Defined structure and clarity	Low

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4. Conclusion

This assignment highlights the importance of well-structured prompt design in Generative AI systems. Poorly designed prompts can lead to unclear, incorrect, or even unsafe responses.

By applying structured improvements such as role definition, clear instructions, logical flow, and security awareness, prompts can produce more accurate, helpful, and reliable outputs.

Effective prompt engineering is essential for building robust and trustworthy AI-based customer support systems.